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Performance Metrics Evaluation Towards The Effectiveness of Data Anonymization

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***Abstract*— A supplementary method for ensuring that private data is inaccessible to outside parties is data anonymization. Anonymization might affect the outcomes of data mining procedures since it may make it more difficult for commonly used algorithms to analyze the data. This practical experience report compares the performance impact of current data anonymization algorithms to the suggested k-anonymization methods utilizing both original and anonymized data in order to assess the correctness and execution time. Through the use of k-anonymization, l-diversity, t-closeness, and differential privacy techniques, a sample of genuine data produced by a healthcare facility was made anonymous. Contrary to predictions, the Hadoop framework was able to handle anonymization approaches, improving accuracy and performance while speeding up execution. These findings show that data anonymization techniques, when properly implemented through Hadoop ecosystems, can help to increase the effectiveness of data anonymization. Furthermore, the suggested method can produce the data anonymization with the necessary utility and protection trade-offs and with a performance scalable to large datasets.**

Keywords: Anonymization, Hadoop, Performance, Data utility, execution time, accuracy.

I. INTRODUCTION

Health care facilities now offer their clients a vast array of web-based services. Major reorganizations and adjustments are being made by healthcare organizations to address the rising demands for better healthcare access, quality, and affordability. Much of the crucial information required to solve these problems is being saved in digital form as the usage of information technology to process medical data rises. The ability to access and integrate healthcare data from many computer programmes and other information sources is a benefit of web-enabled information technology more easily. Users typically have to give private information, like addresses, social security numbers, postal identification number, and health-related data, in order to access these services. Employing data mining techniques, businesses and organizations aim to efficiently gather and analyze this information. Data mining aims to transform unusable data into knowledge that can, for instance, be used to create more efficient marketing strategies, boost sales, and cut costs. There is a chance that private or sensitive data may be disclosed in the process of applying data mining techniques which includes the classification algorithms, that creates privacy concerns. efficiently using data mining techniques.

The use of data anonymization techniques is one of the potential answers to the problem of privacy preservation in data mining applications. Techniques for reducing the disclosure of personal information about a specific person when the data are exchanged and made public are included in data anonymization [1]. In essence, these procedures eliminate or transform individually distinguishable information into aggregated data. But because they might affect the outcomes of data mining, choosing and using these approaches in the context of big data is not an easy undertaking. Finding patterns that identify a patient as having a particular illness may become more difficult or perhaps impossible if some fields in a set of medical data are not carefully adjusted to become anonymous.

This paper presents a study to evaluate the execution time and compare the performance impact of existing data anonymization algorithms to the proposed k-anonymization algorithms.

To perform data anonymization we propose a k-anonymization algorithm for both structured and unstructured data executing Scalable Two-Phase Top-Down Specification approach [2] and NLP approach [3]. We design an appropriate Hadoop MapReduce program to deal with the huge amount of data and also to manage and process data on YARN to perform Job scheduling and Resource management with embedding Python with HDFS to support data processing [4]. We can better assess the utility of the data after anonymization by evaluating accuracy. Understanding the significance of the suggested anonymization technique is made easier by evaluating its performance and contrasting it with that of existing algorithms. Real data from a nearby healthcare facility was used in the experiments.

To even more effectively assess the effects of various anonymization methods individually, we divide the paper in such that we initially discuss the related papers, then brief on the proposed anonymization process followed up with the performance evaluation of the k- anonymization. There were four phases of the performance evaluation. In order to establish the initial comparison parameters, we classified the original table in the first step without any anonymization. In the second stage, we generalize and suppress the original table's quasi-identifier fields to make it anonymous (such as postal code, date of birth, age, etc). Once more, the classifiers received the anonymized data to be evaluated for accuracy and performance. Finally, in the fourth step, we use the k-anonymity model to compare the performance and accuracy of the four stages'

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output. The proposed k- Anonymization algorithm was used to perform the anonymization.

II. RELATED WORK

Remarkably, little study has been done to examine how well data mining algorithms function on anonymized data, according to Inan et al. [5]. Buratovic' et al. [6] evaluated the impact of anonymization by contrasting the original data set with the outcomes of mining the anonymized data set in order to ascertain if anonymized data may be used for research purposes. Data on the students were mined using the WEKA (Waikato environment for Knowledge Analysis) programme after being made anonymous with k-anonymity. Results demonstrate that anonymization can almost completely maintain the integrity of data mining findings while also protecting privacy.

Brickell and Shmatikov [7] used the effectiveness of data-mining algorithms to gauge the adjustment between utility and privacy. In order to quantify attribute disclosure, they created a system based on semantic definitions and measurements. Their findings demonstrate that, in the majority of situations, trivial sanitization which could eliminate all the quasi-identifiers or all the sensitive attributes—offers identical value and superior privacy than models based on generalization and suppression like k-anonymity and l-diversity.

Classification over anonymized data is a problem that is addressed by Inan et al. [5]. They proposed a technique that models could generalize the attributes of anonymized data as unreliable information, where each of the generalized rate of an anonymized record is supplemented by statistics acquired from records belonging to the same equivalence class. Along with the anonymized data, this additional information facilitates the precise estimation of predicted values for data analysis. This method allows for the direct use of locally anonymized data in privacy-preserving distributed data mining without significantly degrading classification accuracy.

Our strategy departs in two keyways. We do not focus on the data semantics since we want to get to anonymization as simple as possible and use the measures provided by the tools used in the tests, once they have already been proven and established in the market. Second, we do not add information to the raw data because doing so would introduce a trend into the data mining activity. Our tests also take into account the important characteristics and sensitive data from the data set in addition to the quasi-identifiers. As a result, we conducted our research using a wider range of attribute types.

Our approach is more in accordance with [8] since we suggest comparing the correctness of original and anonymized data using the tools at our disposal rather than making additional changes to the original data. However, in addition to, our suggested anonymization phases enable us to conduct a more thorough investigation of the effects of each technique separately, including anonymity. In addition to accuracy, we also assessed how the anonymization procedure affected performance.

III. DATA ANONYMIZATION

De-identification, another name for data anonymization, refers to a set of procedures that can be used to stop the recovery of specific user information. A specific individual cannot be described when they are a member of a group of subjects, which is what the anonymous adjective means when it comes to third parties dealing with their data [9]. A record or transaction is termed anonymous when its data, taken alone or in combination with other data, cannot be used to identify a specific person [10].

Anonymization in general is a mechanism to convert an original data into an anonymized form where any exterior information is shared to the private association only on promising to preserve privacy of any data. Anonymization is the progressive method of disturbing the alignment or representation of original data by using some techniques such as generalizing and suppressing the identifiers those which are observed to be a potential threat while linking them with any publicly available databases. The Anonymization algorithms aim at privacy requirements and ensure that the anonymized data remains useful for analysis [11]. Privacy models perform data transformation to achieve privacy requirements, which preserves data utility. Protection of sensitive data from any un-trusted code is always considered to be an imperative security challenge. Magnificently solving this challenge will definitely have the blow-out effect in decreasing security fissures, improving the customer confidence in software products, and stimulating the adoption of the high profitable occupation of cloud computing services.

In order to safeguard an individual's privacy, data can be anonymized using a variety of approaches. These approaches, which are now in use, include generalization, suppression, encryption, and perturbation/masking. The objective of using these techniques is to make the data anonymous. In addition, data anonymization might benefit from models that can be applied to a particular category of data to prevent re-identification, mostly because of privacy attacks, which aim to reveal private user data. These types of privacy preserving techniques considered are k-anonymity, l-diversity, t-closeness, and differential privacy.

The experimental study that was utilized to assess the accuracy and performance impact of the anonymization procedure is presented in the next section.

IV. PERFORMANCE METRICS EVALUATION FOR K-ANONYMIZATION

Since personal information about individuals should not be available in published data sets, big data owners must change the original data before publishing or sharing. Sensitive data alteration reduces data utility, which is the opposite of what should be convenient for maintaining data utility. When releasing data, the original data sets are protected by the privacy-preserving data publishing process, which modifies data for privacy and utility. In order to protect people's privacy, the original data set is anonymized before publication by removing identifiers and altering quasi-identifiers. Relationships between quasi-identifiers and sensitive attributes are broken down during anatomization. By grouping various

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data records and combining the sensitive values in each group, permutation also breaks the link between a quasi-identifier and a sensitive attribute. By switching, introducing noise, or producing fake data, perturbation replaces original values with new ones. These anonymization processes reduce the value of the material, which is reflected in overall information loss. In other words, smaller information loss is associated with increased data usefulness.

Sharing information of any individual with third parties is a matter of privacy concern which can be achieved through the k-anonymization technique. Techniques for data anonymization can be used to limit the disclosure of personal information about a specific person when the data are shared and made public. In a nutshell, these methods eliminate personal identifiers or transform individually identifiable information into aggregated data. Below is a diagram that demonstrates how k-anonymity functions. The tables 1 and 2 display the initial data and the 2-anonymized data, respectively, following the use of the k-anonymization procedure.

TABLE I. ORIGINAL DATA SET

Name	Postcode	Age	Gender	Region	Diseases
*	SE1 ***	20-29	Male	South	Cardiovascular
*	SE1 ***	20-29	Male	South	Respiratory
*	SE1 ***	10-19	Male	West	No illness
*	EW3 ***	40-49	Female	East	Cancer
*	EW3 ***	40-49	Female	West	No illness
*	EW3 ***	50-59	Female	East	Cardiovascular
*	NS6 ***	20-29	Male	North	Respiratory
*	NS6 ***	20-29	Male	North	Liver disorder
*	NS6 ***	10-19	Female	South	Cancer
*	SE1 ***	20-29	Male	South	Liver disorder
*	SE1 ***	20-29	Male	South	Cancer
*	SE1 ***	10-19	Male	West	Cardiovascular
*	EW3 ***	40-49	Female	East	Respiratory
*	EW3 ***	40-49	Female	West	No illness
*	EW3 ***	50-59	Female	East	Cancer
*	NS6 ***	20-29	Male	North	No illness
*	NS6 ***	20-29	Female	South	Cardiovascular
*	NS6 ***	10-19	Female	South	Respiratory

The evaluation metrics used to identify the efficiency of structured data anonymization in the proposed work was in terms of execution time, data utility and complexity.

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A. *Execution time:*

The following table and graph in the figure 1. shows the execution time, the time taken by the different anonymization algorithm to perform the task for different data size.

Fig. 1. Comparing different anonymization algorithms calculating the execution time

B. *Data utility loss and privacy loss:*

The amount of utility that remains in the anonymized data is the subject of the data utility method. This includes data like the average size of the equivalence classes and the discernibility score. This also includes the approach of evaluating the workloads for data mining. Utility can be determined in two different methods, generally. The utility of the anonymized data is measured in the first technique. This includes data like the average size of the equivalence classes and the discernibility score. This also includes the approach of evaluating the workloads for data mining. Calculating the utility loss caused on by data anonymization is part of the second technique. By contrasting the original data with the anonymized data, this is evaluated.

Instead of measuring utility gain, we measure utility loss in comparison to the initial data. The baseline dataset must contain accurate information about huge populations because only accurate information leads to utility. Instead of measuring utility for categorization, we measure it for aggregate statistics. This is due to the fact that numerous studies have demonstrated that when privacy constraints are implemented, the utility of the anonymized data in categorization is constrained. In the worst-case scenario, we quantify privacy loss and total utility loss. Thus, our methodology assesses the utility loss for every piece of helpful knowledge and the privacy loss for every individual as shown in the graphically represented figure 2. Utility loss is plotted. on the y-axis, and privacy loss is plotted on the x-axis

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TABLE II. 2-ANONYMOUS DATA FOR COMBINATION OF AGE, GENDER AND REGION ATTRIBUTES

Name	Postcode	Age	Gender	Region	Diseases
Paul	SE1 222	24	Male	South	Cardiovascular
Soloman	SE1 343	27	Male	South	Respiratory
Richie	SE1 521	19	Male	West	No illness
Tressy	EW3 123	44	Female	East	Cancer
Alice	EW3 325	48	Female	West	No illness
Elissa	EW3 622	59	Female	East	Cardiovascular
Moris	NS6 444	29	Male	North	Respiratory
Gilbert	NS6 245	23	Male	North	Liver disorder
Simman	NS6 366	17	Female	South	Cancer
Patrik	SE1 237	22	Male	West	Liver disorder
Shane	SE1 321	25	Male	South	Cancer
Romy	SE1 441	18	Male	West	Cardiovascular
Tinu	EW3 653	41	Female	East	Respiratory
Alaska	EW3 275	45	Female	West	No illness
Evita	EW3 872	55	Female	West	Cancer
Mehek	NS6 734	27	Male	North	No illness
George	NS6 315	22	Female	South	Cardiovascular
Sweedle	NS6 256	19	Female	South	Respiratory

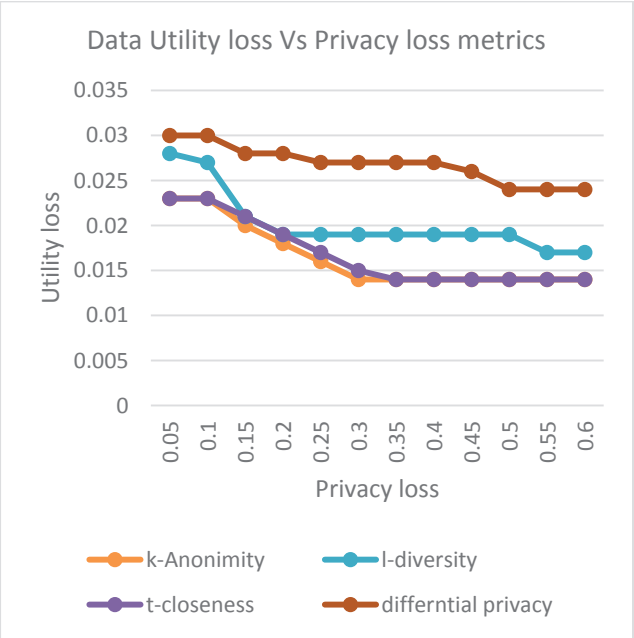

Fig. 2. Data utility loss Vs Privacy loss metrics

In the proposed study, we employ generalization and suppression in anonymization without sacrificing simplification. In other words, the ranges of the group of generalized or suppressed tuples are utilized to express the anonymization when a tuple is generalized or suppressed.

Interpretation of the loss of privacy: We demonstrate the degree of privacy loss quantitatively. We want to know, specifically, what privacy loss an individual would experience if their sensitive value were to be made public.

Analysis of the utility loss: We also give a numerical example of the utility loss. We would like to explicitly address the following issue: What utility loss occurs when all quasi-identifiers are removed? The utility loss is calculated by averaging the utility losses for all substantial populations under the assumption that the predicted distribution is always the overall distribution Q. Our results show that when all quasi-identifiers are removed, the utility loss is 0.05. When compared to publishing trivially anonymized datasets, Figure 2 shows that publishing anonymized data actually improves the quality of the data utility, with utility loss being lower than 0.04 in all circumstances and lower than 0.02 in many instances.

The earlier research demonstrates that under different quality models, the optimal k-anonymity problem is NP-hard. A generalization of the suppression model is the utility-based approach that we provide here. As shown in table 3, we have the following findings about complexity. The definitions can be easily altered to reflect the use of various representations, such as mean or median, and our methods are guaranteed to continue to function properly.

TABLE III. EVALUATE AND COMPARE THE DATA UTILITY LOSS AND PRIVACY LOSS MEASURES FOR VARIOUS MODEL.

S. No.	Models	Data utility loss	Privacy Loss
1	k-anonymity	Low	Very Low
2	l-diversity	Moderate	High
3	t-closeness	low	Moderate
4	Differential privacy	High	Moderate

Accuracy:

Weka checked each experiment's accuracy, i.e., the computation is the same for all sorts of qualities, regardless of their frequency. The accuracy for each class was determined by building a confusion matrix. This matrix shows where the classes' predictions were wrong and where they were right. Because it facilitates the use of machine learning, preprocessing, classification, prediction, and grouping methods, we use the WEKA environment [12] to create the classifiers. The WEKA library's version 3.8 specifies a capacity verification in order to compute the data set prior to building the classifier. In other words, WEKA decides which classifiers can handle input data based on the characteristics of the data set as we feed it into it. The classifiers we utilized for comparison, based on this WEKA selection, are Navie Bayes, J48, Neural Networks, and SVM for various degrees of k values. The information can be processed using classifiers. The classifiers we utilized for comparison, based on this WEKA selection, are Navie Bayes, J48, Neural Networks, and SVM for various degrees of k values.

The Naive Bayes classifier is a statistical classifier that can estimate the likelihood that a sample will belong to a particular class, making it extremely helpful for very large data sets.

Experiments are conducted for different levels of k-anonymity in for different values from (1-200). The anonymized data is classified using Naïve Bayes classifier. The results for classification are also calculated in terms of their precision and recall.

The figure 3 shows that the accuracy decreases with the increase in the k- anonymity while performing the accuracy check for different levels of k values. Figure 4 show precision and recall for different levels of k-anonymity.

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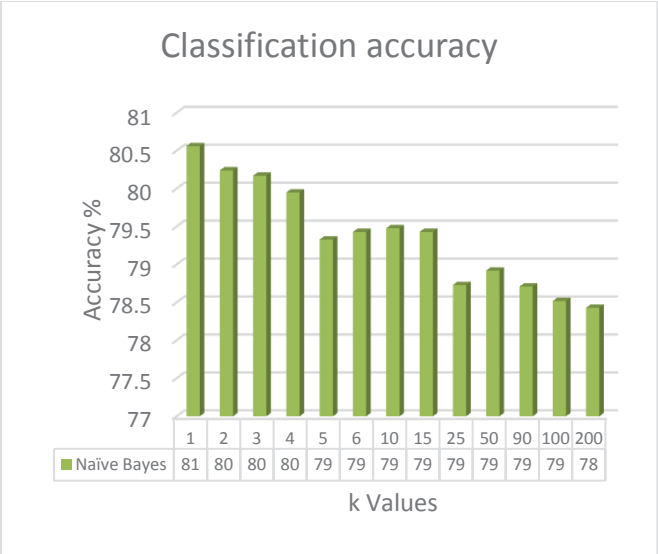

Fig. 3. Classification accuracy for different k values

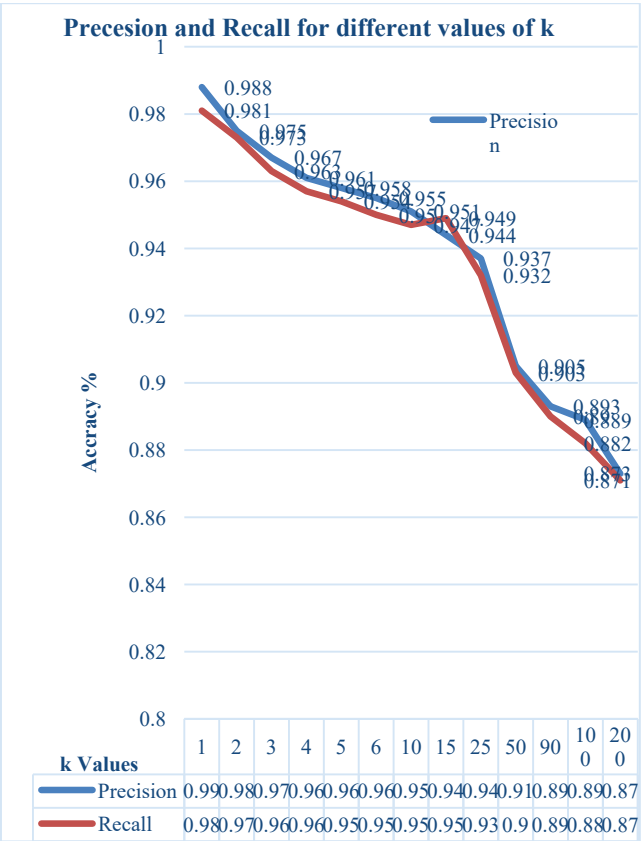

Fig. 4. Precision and recall for different values of k

The observations made from the experimental results indicates that with the increase in the anonymity the performance of the classifiers degrades proportionately.

V. CONCLUSION

We were able to extend the trials described in earlier studies for a wider variety of attribute types thanks to the anonymization in phases, which improved our ability to assess the impact of anonymizing various attribute kinds. These earlier studies' findings are supported by our results, which show that generally speaking, using anonymization approaches has no appreciable effects on the performance and accuracy of classifiers. The accuracy of the classifiers tends to rise when anonymization approaches like generalization are utilized. When looking at the data output, it is necessary to delete the sensitive attributes more cautiously so that the classification does not lose its value as a result of their suppression. When employed in the preprocessing and data sanitization stage of data mining, anonymization can help with the classification stage. By using the anonymity method and analyzing the data mining results, we can increase the performance and accuracy of the classifiers while also protecting the privacy of personally identifiable information.

We trusted the WEKA platform's results for the experiments because it offered the accuracy and performance measurements for this investigation. In order to expand the research, we want to test the same assortment of attribute types using different data sets (such as clinical data sets) in future work.

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